

Terminology and Mathematical Language: Worked Solutions

Problem 1. A prediction and an action

Problem. A program predicts a probability of rain and recommends whether to carry an umbrella. Identify the task, prediction, action, model parameters, and a possible source of experience. Which of these changes if carrying an umbrella becomes more costly?

Solution. Let x describe the available weather measurements and let $Y \in \{0, 1\}$ indicate rain during the specified forecast period. The prediction task is to estimate $P(Y = 1 \mid x)$. Its output is a number $q_\theta(x) \in [0, 1]$. The decision task is to choose an action $a \in \{0, 1\}$, where one means carrying an umbrella. A probability and an action are different outputs: the same probability can lead to different actions when their consequences differ.

For example, take

$$q_\theta(x) = \frac{1}{1 + \exp[-(w^T x + b)]}, \quad \theta = (w, b).$$

The entries of w and the intercept b are the model parameters. Historical pairs of weather measurements and subsequent rain observations are a possible source of experience. A learning algorithm uses these pairs and a chosen loss to fit θ . Evaluating the resulting $q_\theta(x)$ on today's measurements produces the probability prediction; applying a cost rule to that probability produces the recommendation.

To make the cost question precise, suppose carrying an umbrella costs $c_U \geq 0$ whether or not it rains, whereas getting caught in the rain without one costs $c_R > 0$. Assume an umbrella avoids that latter cost. The conditional expected costs are

$$\mathbb{E}[C(1, Y) \mid x] = c_U, \quad \mathbb{E}[C(0, Y) \mid x] = q_\theta(x)c_R.$$

Thus carrying is optimal when $q_\theta(x) \geq c_U/c_R$, with either action optimal at a tie. Increasing c_U raises the threshold. If $c_U > c_R$, carrying is never optimal under this model. The decision criterion and possibly the action change; the observed experience, fitted weather parameters, and probability prediction need not change. A different cost model can give a different threshold, so the costs must be specified before a numerical action rule can be deduced.

Problem 2. Observed errors and population risk

Problem. A fixed function makes errors on 12 of 80 independent test cases. State the empirical zero-one risk, its population counterpart, and what extra assumptions would be needed to describe the uncertainty of the estimate. Why does choosing the best of many functions on those same 80 cases change the interpretation?

Solution. For a classifier f , write $E_i = \mathbf{1}\{f(X_i) \neq Y_i\}$. The observed empirical risk is

$$\widehat{R}_D(f) = \frac{1}{80} \sum_{i=1}^{80} E_i = \frac{12}{80} = 0.15.$$

Its population counterpart under the intended test distribution P is

$$R_P(f) = \mathbb{E}_{(X,Y) \sim P}[\mathbf{1}\{f(X) \neq Y\}] = P\{f(X) \neq Y\}.$$

The observed proportion is known; the population probability is not determined by the error count alone.

For the usual binomial uncertainty calculation, assume the cases are identically distributed according to P as well as independent, and that f was fixed independently of these test observations. Conditional on that fixed function, the errors are then iid Bernoulli with parameter $p = R_P(f)$. Consequently,

$$80\hat{R}_D(f) \sim \text{Binomial}(80, p), \quad \mathbb{E}[\hat{R}_D(f)] = p, \quad \text{Var}(\hat{R}_D(f)) = \frac{p(1-p)}{80}.$$

The plug-in standard error is $\sqrt{0.15(0.85)/80} \approx 0.0399$. It estimates sampling variability under these assumptions; it is not itself a confidence interval. A binomial interval procedure would also require a chosen confidence level. Independence without a common error probability does not imply the displayed binomial law, and a shifted test distribution does not measure risk under the original P .

If the same 80 cases select the lowest-error function from a collection, the selected function depends on the test outcomes. Favorable random fluctuations influence both its selection and its reported loss. The fixed-function binomial calculation therefore cannot simply be applied after selection, and the minimum observed error is generally an optimistic performance estimate. Evaluating the selected function on fresh independent cases restores the fixed-function interpretation conditional on the completed selection process. Alternatively, a simultaneous bound can explicitly account for the collection being searched.

Problem 3. Shapes in the chain rule

Problem. Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ and $L : \mathbb{R}^2 \rightarrow \mathbb{R}$. Write the shapes of J_f , $D_y L$, $D_x(L \circ f)$, and $\nabla_x(L \circ f)$. Explain why $D_y L J_f$ is a valid product.

Solution. Put $y = f(x)$ and assume both functions are differentiable at the points in question. In numerator layout, output coordinates index rows and input coordinates index columns. Hence

$$J_f(x) \in \mathbb{R}^{2 \times 3}, \quad D_y L(f(x)) \in \mathbb{R}^{1 \times 2}.$$

The differential identities $dy = J_f(x) dx$ and $dL = D_y L dy$ give

$$D_x(L \circ f)(x) = D_y L(f(x)) J_f(x) \in \mathbb{R}^{1 \times 3}.$$

The inner dimensions of the product are both two: one sums over the intermediate coordinates y_1, y_2 . In coordinates,

$$\frac{\partial(L \circ f)}{\partial x_j} = \sum_{i=1}^2 \frac{\partial L}{\partial y_i} \frac{\partial f_i}{\partial x_j}.$$

The Euclidean gradient is the transpose of the scalar row derivative, so

$$\nabla_x(L \circ f)(x) = J_f(x)^T \nabla_y L(f(x)) \in \mathbb{R}^{3 \times 1}.$$

The row derivative acts on an input displacement; its transpose is the column used in the Euclidean inner product with that displacement.

Problem 4. A self-supervised sequence task

Problem. Describe a self-supervised task on a sequence without manually supplied labels. Identify the input and target of each training example, and distinguish the learning objective from the model architecture.

Solution. For an observed token sequence $(w_1, \dots, w_T)$, form examples

$$x_t = (w_1, \dots, w_{t-1}), \quad y_t = w_t, \quad t = 2, \dots, T.$$

The input is a prefix and the target is its next observed token. The targets are obtained from the sequence itself; no separate human label is required. The input excludes the target token, so the task requires prediction rather than copying an exposed answer.

A model returns a distribution $p_\theta(\cdot \mid x_t)$ over possible next tokens. For $T \geq 2$, one possible fitting objective is the average negative log likelihood

$$\hat{R}(\theta) = -\frac{1}{T-1} \sum_{t=2}^T \log p_\theta(w_t \mid w_1, \dots, w_{t-1}).$$

The objective specifies how a proposed conditional distribution is scored. The architecture specifies how the parameters compute that distribution: an RNN or a causal Transformer could realize the same prediction task and loss. The learning algorithm specifies how the parameters are obtained, for example by stochastic gradient updates. These are separate choices. Finally, overlapping prefixes from one sequence are dependent examples; the construction does not make them iid.